

# PRANAY KUMAR

AI Researcher | ML/DL | Multi Modal AI

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## PROFILE

*I'm a 4th-year Integrated M.Tech student who genuinely enjoys digging into how AI systems break — and how to make them better. Most of my research has revolved around adversarial robustness and the security of Vision-Language Models and DNNs, work I've had the chance to do at IISER Bhopal and IIT Bhubaneswar, which got me thinking a lot about how these models behave in the real world. Currently exploring Multi-Modals and Generative Models — always chasing the next hard problem where trustworthy AI still falls short.*

## EDUCATION

**Integrated M.Tech in INFORMATION TECHNOLOGY** Aug 2023 – Present

International Institute of Professional Studies, DAVV, Indore | CGPA: 7.64 / 10

**Senior Secondary (12th Board)** Apr 2023

Score: 80.4 / 100

## RESEARCH & PROJECTS

**AI Researcher** Present

*IISER Bhopal*

- Working on security and robustness of Vision-Language Models (VLMs)

**Research Intern** Dec 2025 – Feb 2026

*IIT Bhubaneswar (Remotely)*

- Worked on an LLM-based Trip Planner in collaboration with Microsoft India
- Designed and created structured trip plans used for model evaluation and benchmarking

**Research Intern** May 2025 – Jul 2025

*IISER Bhopal*

- Conducted experiments based on the ICLR 2025 paper: Democratic Training Against Universal Adversarial Perturbations
- Fine-tuned deep learning models to enhance adversarial robustness
- Implemented entropy regularization and inter-layer agreement techniques for improved adversarial defense

## SKILLS

**Programming:** Python, C++, MySQL

**Frameworks & Libraries:** PyTorch, NumPy, Pandas, Matplotlib

**AI/ML Domains:** Deep Learning, Computer Vision, Transformers, LLMs, Vision-Language Models (VLMs), Adversarial Robustness

**Other:** Generative AI, Social Network Analysis, Data Structures

## NPTEL COURSEWORK

- Deep Learning for Computer Vision
- Introduction to Large Language Models (LLMs)

- Programming with Generative AI
- Social Network Analysis

PRESENTATIONS & WORKSHOPS

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| <b>Participant &amp; Presenter</b>                                                                                                                                                                                                  | <i>July 2025</i>  |
| <ul style="list-style-type: none"><li>• Presented research on adversarial robustness and democratic training in AI/ML systems</li><li>• Participated in workshops on AI ethics, deep learning, LLMs, and interpretability</li></ul> |                   |
| <b>Workshop</b>                                                                                                                                                                                                                     | <i>June 2025</i>  |
| <ul style="list-style-type: none"><li>• Explored data forgetting techniques and ethical model design for generative AI systems</li></ul>                                                                                            |                   |
| <b>Symposium</b>                                                                                                                                                                                                                    | <i>March 2025</i> |
| <ul style="list-style-type: none"><li>• Attended sessions on AI-driven diagnostics, medical imaging, and healthcare innovation</li></ul>                                                                                            |                   |

LANGUAGES

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English (Professional), Hindi (Native)